

In Figure 1, a simplified map shows 7 cities connected by directional roads, each associated with a positive integer travel time. A robot plans to start its journey from city S using *Graph Search* variant of search algorithms to find a path to the goal. Assume that a tie in the fringe is broken randomly. There are three possible goal nodes, namely G_1 , G_2 , and G_3 .

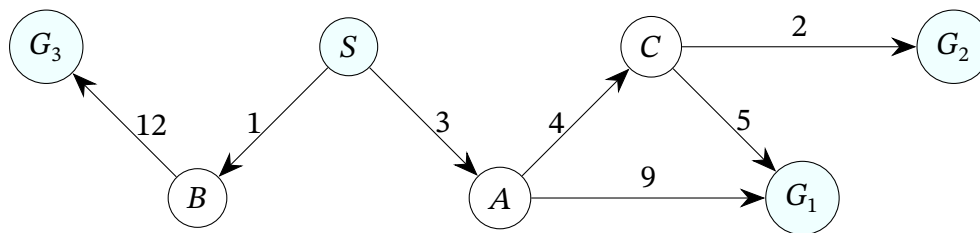


Figure 1: A simplified map of a country

Three heuristics, h_1 , h_2 , and h_3 are shown in Table 1.

Table 1: Heuristics Functions

	A	B	C	S	G_1	G_2	G_3
h_1	0	0	0	0	0	0	0
h_2	6	7	1	7	0	0	0
h_3	7	7	1	7	0	0	0

Based on the aforementioned information, answer the following questions:

1. State the branching factor and the maximum depth of the search tree.

Solution:

$b = 2$ and $m = 3$.

Rubric:

- 0.5 points for each.

1
(CO1)
(PO1)

2. With a brief justification, state the possible goal node(s) that can be returned by each of the following algorithms:

11
(CO1)
(PO1)

- a) Depth-First Search (DFS)

Solution:

G_1 , G_2 , and G_3 . DFS can arbitrarily break ties on the fringe, so the search can reach any goal.

Rubric:

- 0.5 points for the correct ones.
- 1 point for the justification.

- b) Breadth-First Search (BFS)

Solution:

G_1 and G_3 . BFS will only find the goals on the shallowest depth.

Rubric:

- 0.5 points for the correct ones.
- 0.5 points for omitting the wrong one.
- 1 point for the justification.

c) Uniform Cost Search (UCS)

Solution:

G_2 . UCS will find the closest goal based on distance.

Rubric:

- 0.5 points for the correct one.
- 0.5 points for omitting the wrong ones.
- 1 point for the justification.

d) Greedy Search with h_1

Solution:

G_1 , G_2 , and G_3 . Greedy search with trivial heuristic is equivalent to DFS.

Rubric:

- 0.5 points for the correct ones.
- 1 point for the justification.

e) Greedy Search with h_3

Solution:

G_1 and G_3 . Greedy search will either explore node A or B and find the goal immediately as their heuristic value is 0.

Rubric:

- 0.5 points for the correct one.
- 0.5 points for omitting the wrong ones.
- 1 point for the justification.

f) A* Search with h_1

Solution:

G_2 . A* with trivial heuristic is equivalent to UCS.

Rubric:

- 0.5 points for the correct one.
- 0.5 points for omitting the wrong ones.
- 1 point for the justification.

3. For each of the three heuristics shown in Table 1, indicate whether it is consistent, admissible, or neither.

3
(CO1)
(PO1)

Solution:

h_1 is consistent and admissible. (Trivial heuristics are always both)

h_2 is admissible, but not consistent. (Since $h_2(C) - h_2(A) > cost(A, C)$)

h_3 is neither. ($h_3(A) > \text{Optimal cost of } A \text{ to } G_2$)

Rubric:

- 1 point for each.
- (You do not have to write the explanation, as it was not asked. They are provided here for your understanding purpose.)